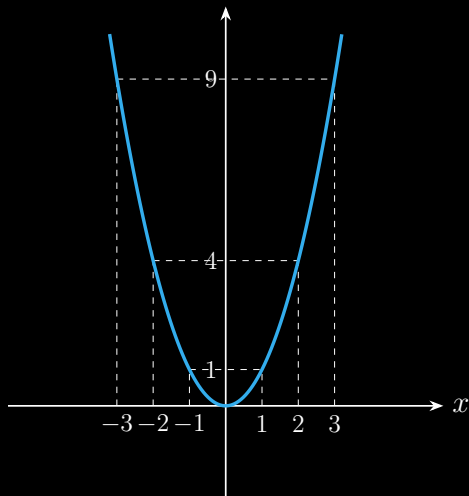
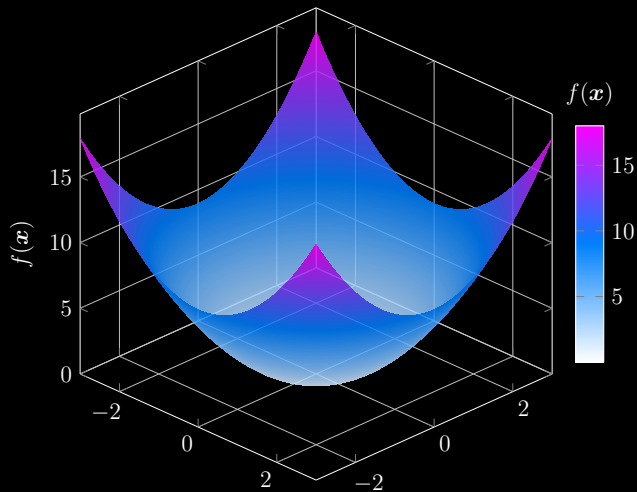


Quadratic Functions  $f(x) = ax^2$  vs.  $f(x) = x^T Ax$



Positive  $a > 0$



Positive definite  $A \succ 0$

## Positive Semidefinite Matrix $\mathbf{A} \succeq 0$

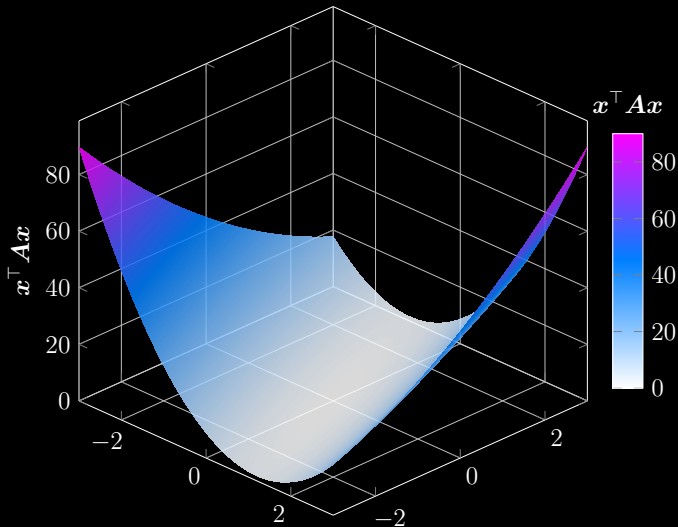
$\mathbf{A}$  is a positive semidefinite matrix if and only if

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$$

for any  $\mathbf{x} \in \mathbb{R}^n$ .

Example:

$$\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} \succeq 0$$



## Positive Definite Matrix $\mathbf{A} \succ 0$

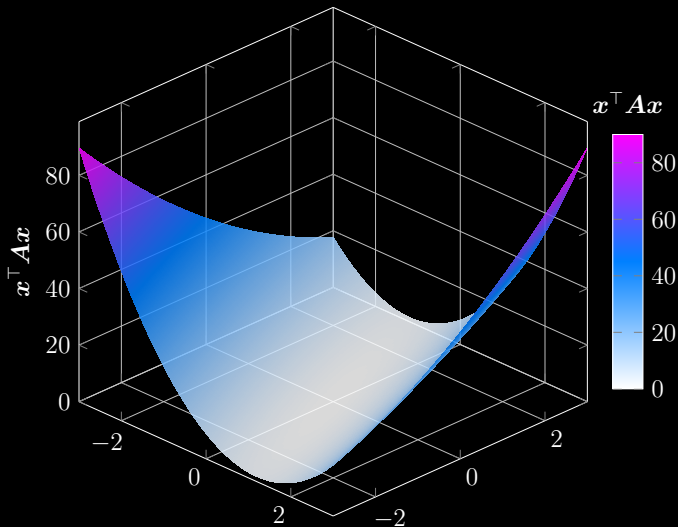
$\mathbf{A}$  is a positive definite matrix if and only if

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$$

for any nonzero vector  $\mathbf{x} \in \mathbb{R}^n$ .

Example:

$$\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} \succ 0$$



Positive Definite Matrix  $A = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$

Eigenvalue decomposition:

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 5.83 \quad \lambda_2 = 0.17$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.92 \\ 0.38 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.38 \\ 0.92 \end{bmatrix}$$

```
1 import numpy as np
2
3 A = np.array([[5, 2], [2, 1]])
4 eig_val, eig_vec = np.linalg.eig(A)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```

# Ellipsoid Representation

Representing positive definite matrices as ellipsoids:

$$\mathcal{E}_A = \{\mathbf{x} \mid \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x} \leq 1\} \quad (\text{centered at the origin})$$

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Inverse of eigenvalue decomposition:

$$\begin{aligned} \mathbf{A} &= \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top \\ \Rightarrow \mathbf{A}^{-1} &= \frac{\mathbf{u}_1 \mathbf{u}_1^\top}{\lambda_1} + \frac{\mathbf{u}_2 \mathbf{u}_2^\top}{\lambda_2} \end{aligned}$$

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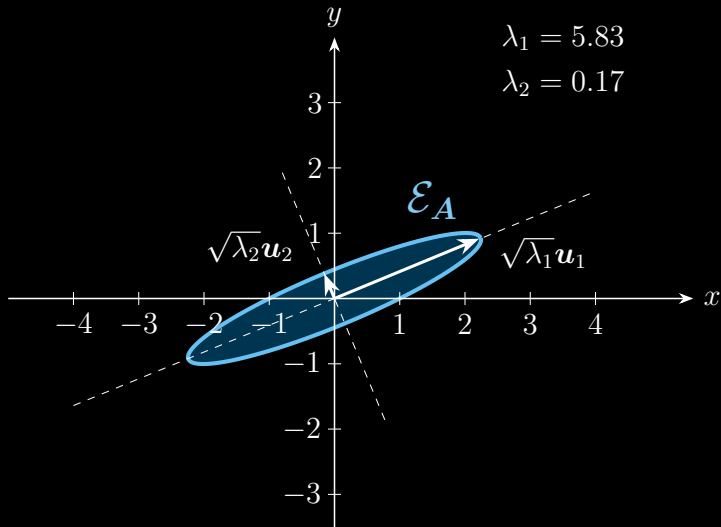
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Geometric interpretation of  $\mathcal{E}_A$ :

$$\mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x} = \frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \frac{(\mathbf{u}_2^\top \mathbf{x})^2}{\lambda_2} \leq 1 \quad \rightarrow \quad \text{ellipse!}$$

Ellipse  $\mathcal{E}_A = \{x \mid x^\top A^{-1} x \leq 1\}$



Positive Definite Matrix  $B = \begin{bmatrix} 10 & 3 \\ 3 & 2 \end{bmatrix}$

Eigenvalue decomposition:

$$B = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

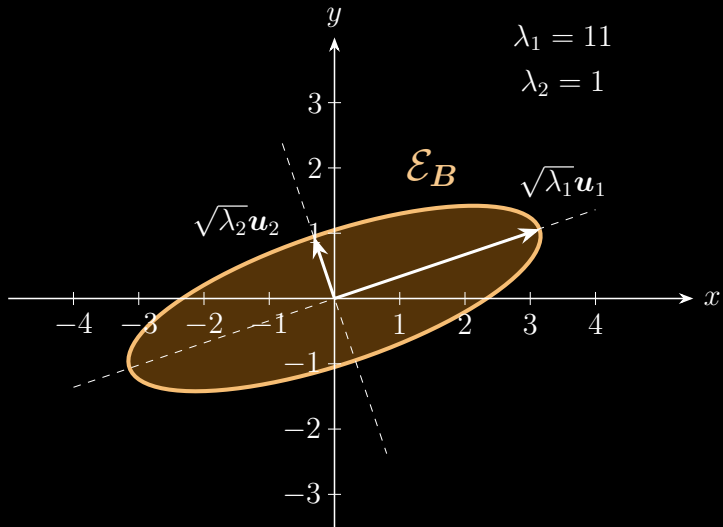
$$\lambda_1 = 11 \quad \lambda_2 = 1$$

and eigenvectors:

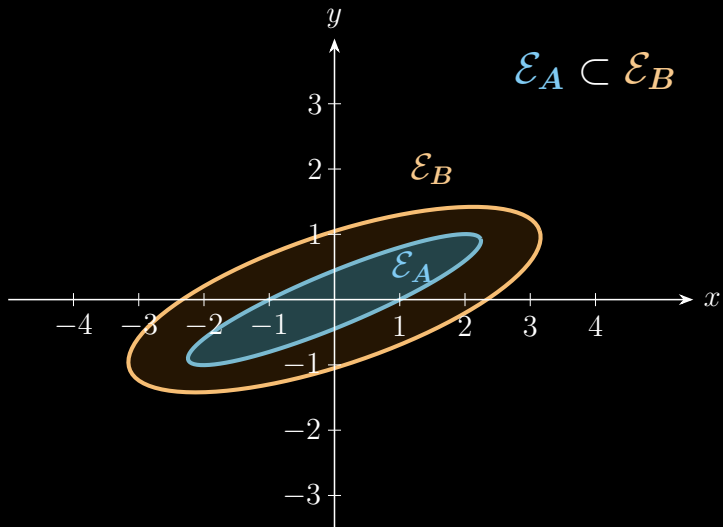
$$\mathbf{u}_1 = \begin{bmatrix} 0.95 \\ 0.32 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.32 \\ 0.95 \end{bmatrix}$$

```
1 import numpy as np
2
3 B = np.array([[10, 3], [3, 2]])
4 eig_val, eig_vec = np.linalg.eig(B)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```

Ellipse  $\mathcal{E}_B = \{x \mid x^\top B^{-1} x \leq 1\}$



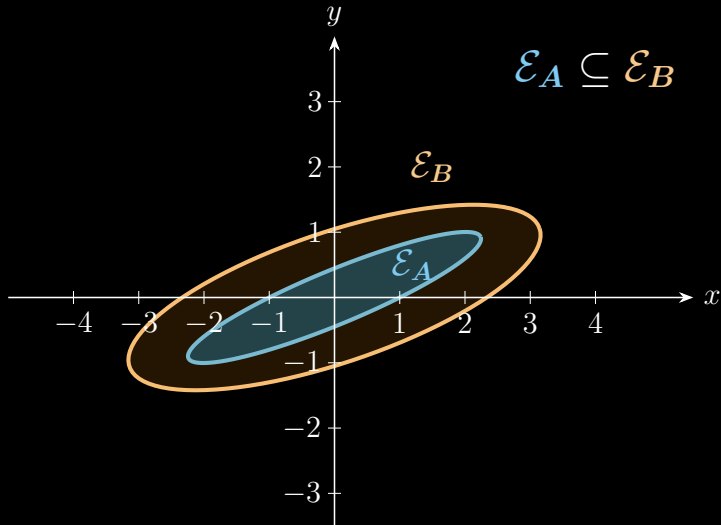
# Positive Semidefinite Ordering



## Positive Semidefinite Ordering

$$\mathcal{E}_A \subseteq \mathcal{E}_B \quad \text{if and only if} \quad \underbrace{B - A \succeq 0}_{\text{positive semidefinite matrix}} \quad \Leftrightarrow \quad B \succeq A$$

$$B - A = \begin{bmatrix} 10 & 3 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 1 \end{bmatrix} \succcurlyeq 0 \quad \Rightarrow \quad \textcolor{red}{B} \succcurlyeq \textcolor{red}{A}$$



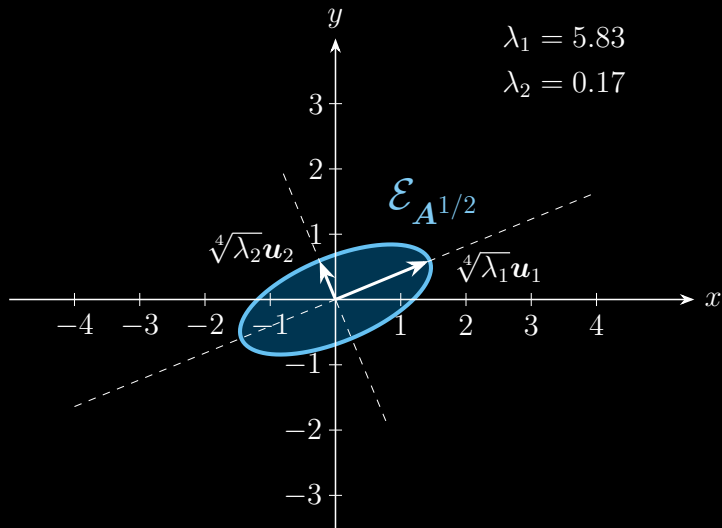
Matrix Monotone Function  $f(\mathbf{A}) = \mathbf{A}^{1/2}$

Ellipse representation:

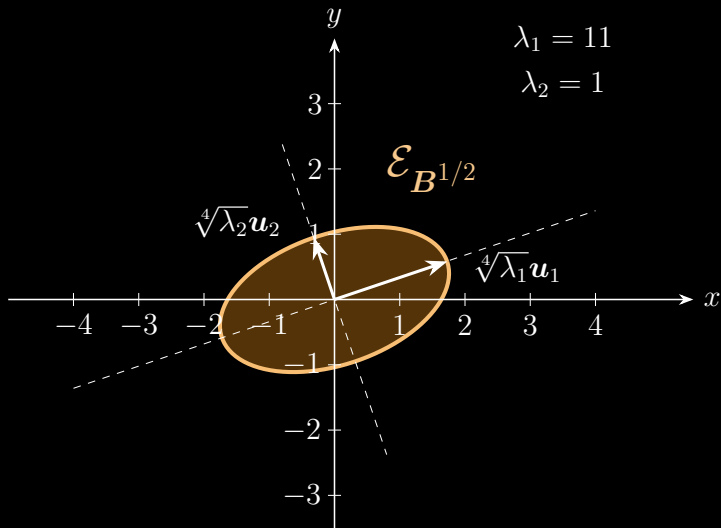
$$\mathcal{E}_{\mathbf{A}^{1/2}} = \left\{ \mathbf{x} \mid \mathbf{x}^\top (\mathbf{A}^{1/2})^{-1} \mathbf{x} \leq 1 \right\}$$

$$\Rightarrow \frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\sqrt{\lambda_1}} + \frac{(\mathbf{u}_2^\top \mathbf{x})^2}{\sqrt{\lambda_2}} \leq 1$$

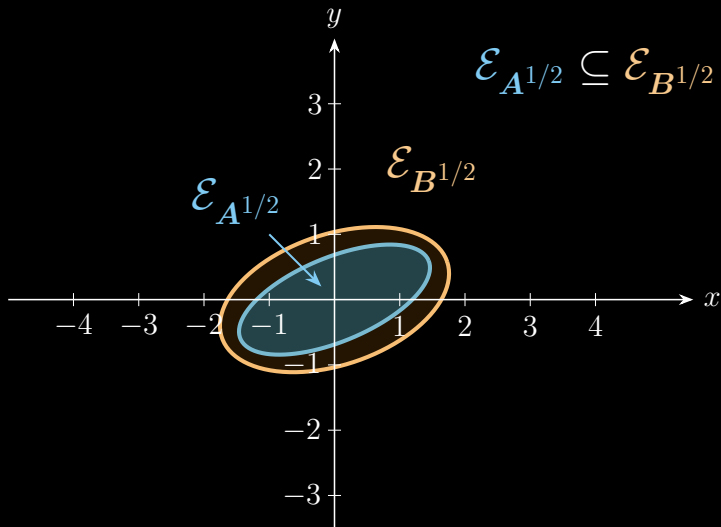
Matrix Monotone Function  $f(\mathbf{A}) = \mathbf{A}^{1/2}$



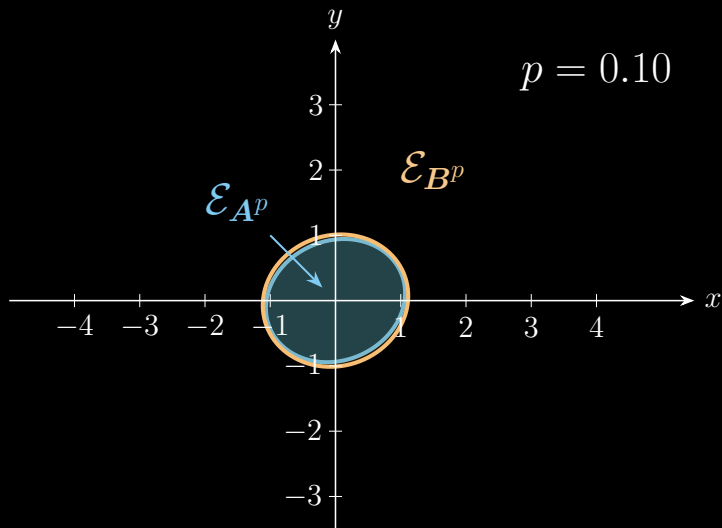
Matrix Monotone Function  $f(\mathbf{B}) = \mathbf{B}^{1/2}$



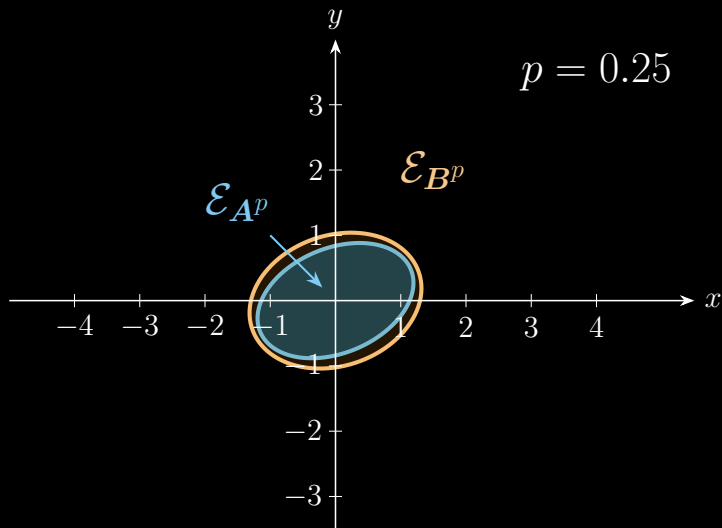
# Positive Semidefinite Ordering $B^{1/2} \succeq A^{1/2}$



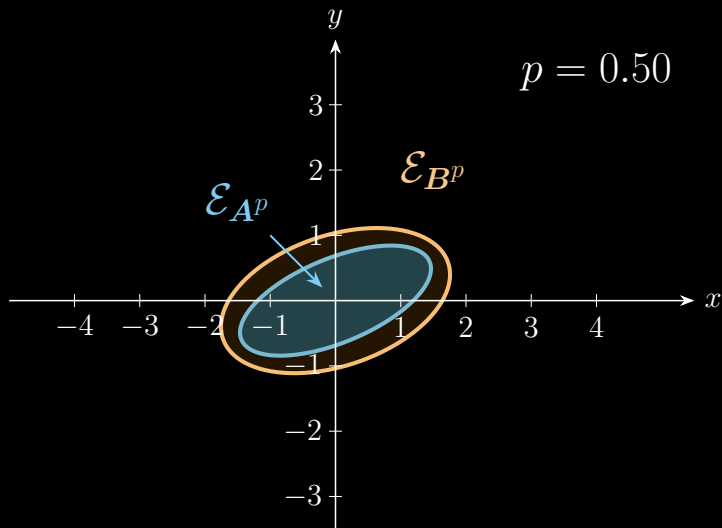
Positive Semidefinite Ordering  $\mathbf{B}^p \succeq \mathbf{A}^p$  for  $0 \leq p \leq 1$



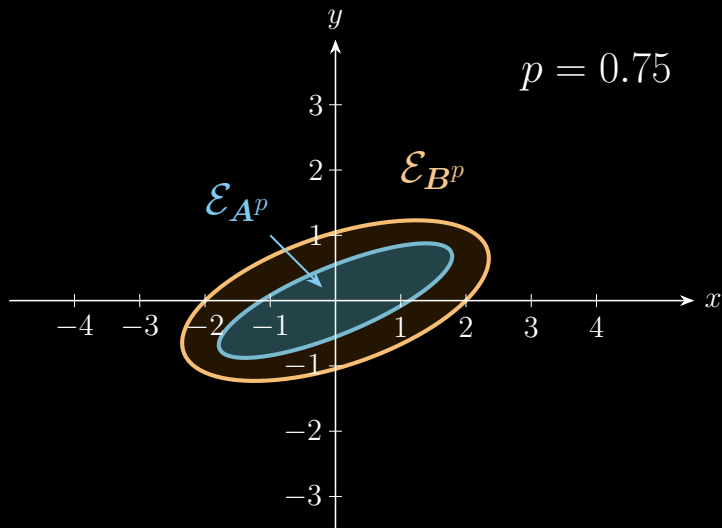
Positive Semidefinite Ordering  $\mathbf{B}^p \succeq \mathbf{A}^p$  for  $0 \leq p \leq 1$



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Positive Semidefinite Ordering  $\mathbf{B}^p \succeq \mathbf{A}^p$  for  $0 \leq p \leq 1$

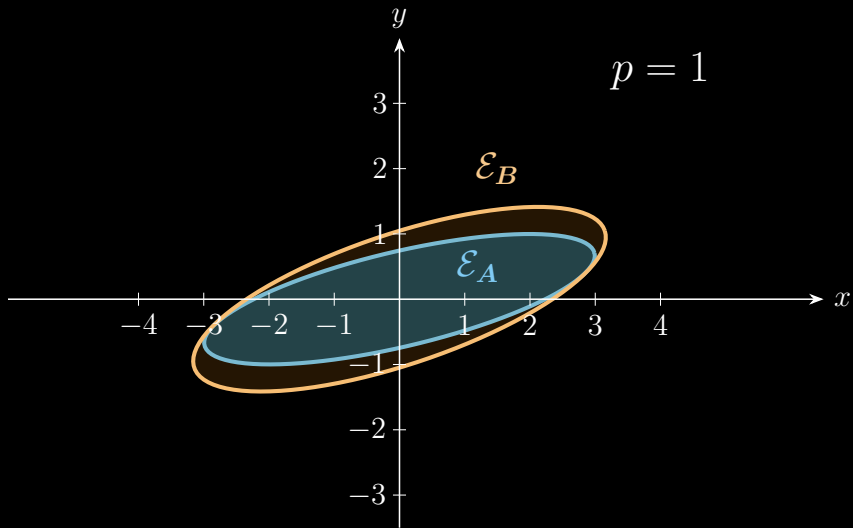


How about  $\mathbf{B}^p \succeq \mathbf{A}^p$  ( $p > 1$ )?

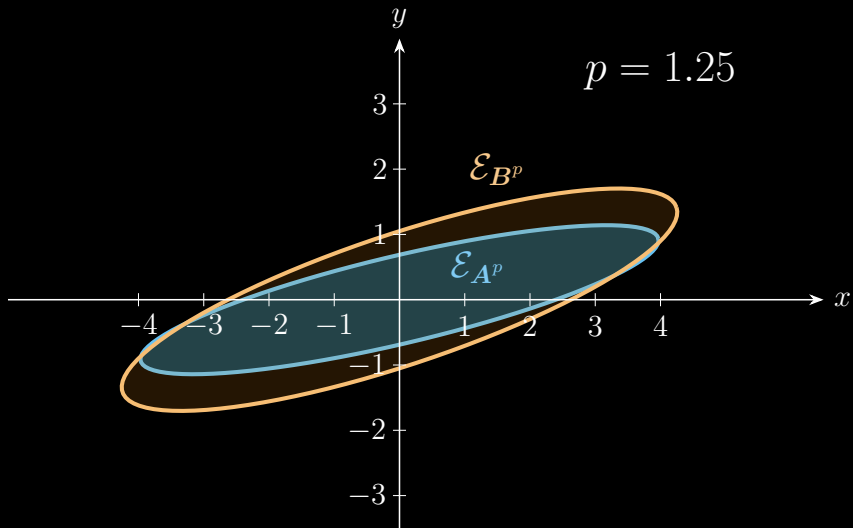
A special case:

$$\mathbf{A} = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 10 & 3 \\ 3 & 2 \end{bmatrix}$$

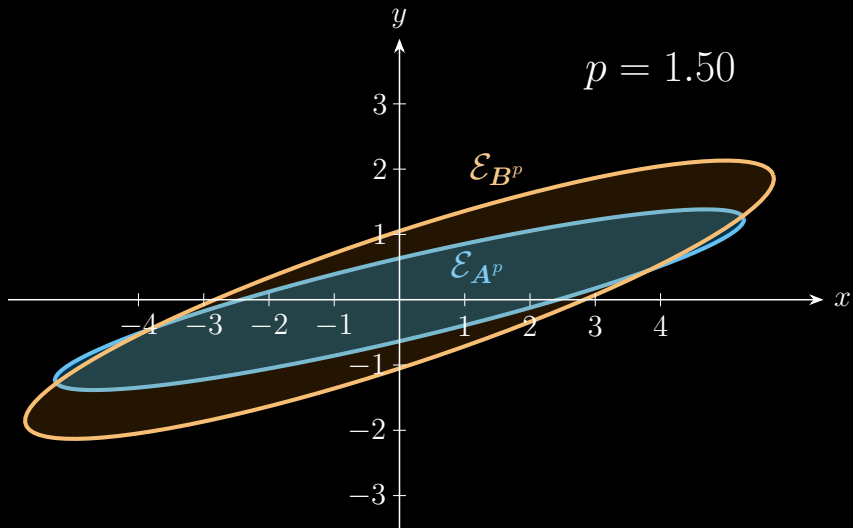
$$B \succeq A \Rightarrow \mathcal{E}_A \subseteq \mathcal{E}_B$$



$$B \succeq A, \text{ but } \mathcal{E}_{A^p} \not\subseteq \mathcal{E}_{B^p}$$



$B \succeq A$ , but  $\mathcal{E}_{A^p} \not\subseteq \mathcal{E}_{B^p}$



## Positive Semidefinite Ordering

For any matrix monotone function  $f(\cdot)$ , it always holds that

$$\mathbf{B} \succeq \mathbf{A} \quad \Rightarrow \quad f(\mathbf{B}) \succeq f(\mathbf{A})$$

For instance,  $\mathbf{B}^p \succeq \mathbf{A}^p$  for  $0 \leq p \leq 1$ .

# Thanks for your attention!

## About me:

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